

Computing Discipline Skills Inventory

Comprehensive extraction of Core Competencies, Employability Skills, and 21st Century Skills across all Computing disciplines from the Nigerian University System (CCMAS, 2022).

Overview: This document serves as a specialized directory containing only the specific skills and competencies detailed in the Core Curriculum and Minimum Academic Standards (CCMAS) for the Computing discipline. It encompasses key qualifications, practical tools, soft skills, and strategic competencies required of graduates across seven undergraduate computing programs.

1. B.Sc. Computer Science

EMPLOYABILITY SKILLS

Graduates of Computer Science are structured to acquire balanced professional and leadership skills to maintain high competitive advantages in the job market:

- **Communication & Teamwork:** Highly articulate verbal and written communication, structured technical briefing, and active cross-functional collaboration.
- **Organisation & Management:** Task planning, structural prioritisation, and technical project lifecycle coordination.
- **Critical Thinking:** Deep technical debugging, algorithmic analysis, and structured logic application.
- **Leadership & Initiative:** Self-directed development, technical mentorship, and proactive technological application.
- **Technology Skills:** Advanced core computing competencies and multi-paradigm software engineering.
- **Self-Management:** Structured time management, target planning, and lifelong professional learning.

21ST CENTURY SKILLS

NUC Standard Skills

- Creativity & Innovation
- Critical Thinking & Complex Problem Solving
- Information Literacy
- Media Literacy
- Flexibility & Adaptability
- Social Skills & Cultural Awareness
- Collaboration & Team dynamics
- Global Awareness

2. B.Sc. Cybersecurity

EMPLOYABILITY SKILLS

Cybersecurity competencies are structured as hard technical capabilities paired with strong presentation and communication systems:

- **Soft Skills:** Excellent presentation and communication skills, ability to clearly articulate complex cyber-concepts, and active listening.
- **Technical Architecture:** Systems administration, virtualization, and configuration of operating systems, enterprise networks, and cloud platforms.
- **Network Defense:** Professional administration of enterprise firewalls, network load balancers, and intrusion detection/prevention systems.
- **Software & Security Analytics:** Software development concepts, analytical tools, and common security scripting languages.
- **Threat Management:** Cyber-threat hunting, active cyber-intelligence, cyber-threat modeling, and vulnerability assessment.
- **Automation & Scripting:** Coding skills to automate defensive operations and standard security compliance tasks.

21ST CENTURY SKILLS

NUC Standard Skills

- Complex Problem-Solving
- Critical Thinking
- Presentation & Communication
- Creativity & Innovation
- Social Skills
- Information Literacy & Technical Analysis
- Collaboration & Cross-functional communication
- Global Awareness

3. B.Sc. Data Science

EMPLOYABILITY SKILLS

Geared towards imbuing students with computational, mathematical, statistical, and analytical competencies to harvest data-driven intelligence:

- **Computational & Statistical Thinking:** Complex algorithm development and statistical inference application.
- **Mathematical Foundations:** Rigorous linear algebra, mathematical modeling, and analytical optimization.
- **Model Building & Assessment:** Designing, evaluating, and refining predictive and descriptive machine learning models.
- **Data Curation & Information Management:** Archiving, cleansing, structural integration, and complete lifecycle governance of large-scale datasets.
- **Knowledge Transference:** Translating raw data-driven insights into actionable business and strategic decisions.
- **Research & Digital Literacy:** Systematic scientific methodology and expert application of modern analytics tools.

21ST CENTURY SKILLS

NUC Standard Skills

- Critical Thinking & Data Analysis
- Creativity & Scientific Curiosity
- Problem Solving & Decision Support
- Information & Media Literacy
- Flexibility & Lifelong Adaptability
- Responsible Digital Citizenship
- Social Skills & Professional Communication
- Global Awareness & Trend Tracking
- Innovation & Entrepreneurial Thinking

4. B.Sc. Information & Communication Technology

EMPLOYABILITY SKILLS

Engineered to prepare graduates for ICT Commercialisation (economic growth enablement) and Technology Management (stable operation environments):

- **Negotiation & Persuasion:** Engaging with stakeholders and presenting commercially viable technology options.
- **ICT Commercialisation:** Packaging technology tools as marketable, scalable, and economically valuable resources.
- **Technology Management:** Deploying, configuring, and maintaining stable operational networks and communications platforms.
- **Organization & Teamwork:** High-stress organization, team leading, collaborative system troubleshooting, and organizational agility.
- **Stress Tolerance & Perseverance:** Exceptional capability to perform and coordinate under intensive operational pressures.

21ST CENTURY SKILLS

NUC Standard Skills

- Critical Thinking & Analytics
- Creativity & Commercial Innovation
- Information & Media Literacy
- Flexibility & Technical Adaptability
- Social Skills & Cross-cultural collaboration
- Problem Solving under operational pressure
- Global Awareness & Digital Trend Literacy

5. B.Sc. Information Systems

EMPLOYABILITY SKILLS

Focuses on managing information system resources and delivering high-quality intelligence for fast corporate and operational decision-making:

- **Information Organisation & Retrieval:** Structured database design, query optimization, and technical asset archiving.
- **Data Analysis:** Analyzing business flows and identifying structural operational patterns.
- **Business Strategy Alignment:** Translating technology assets into strategic corporate advantages and market solutions.
- **System Security Analysis:** Analyzing and identifying potential security vulnerabilities within enterprise information infrastructure.
- **Communication & Presentation:** Presenting highly structured reports to board-level and management stakeholders.

21ST CENTURY SKILLS

NUC Standard Skills

- Creativity & Solution Engineering
- Information Literacy & Asset Valuation
- Flexibility & Technology Alignment
- Social Skills & Human-to-Computer Interaction
- Structured Problem Solving
- Collaboration & Stakeholder management
- Global Business Awareness
- Innovation Skills
- Critical Thinking

6. B.Sc. Information Technology

EMPLOYABILITY SKILLS

Aligns direct operational technology management with structural organizational support to solve day-to-day enterprise user challenges:

- **Problem-Solving & Troubleshooting:** Fast diagnostic capabilities across complex software, networks, and communication setups.
- **Negotiation & Persuasion:** Procuring technology systems and articulating tech value arguments to key deciders.
- **Teamwork & Collaboration:** Operating closely with end-users, design teams, and system integration specialists.
- **Enterprise Organisation:** Structural task prioritisation, asset management, and proactive workflow automation.
- **Stress Tolerance:** Ability to work under high pressure during unexpected enterprise network downtime.

21ST CENTURY SKILLS

NUC Standard Skills

- Creativity & Implementation Engineering
- Information Literacy & Technical Asset Valuation
- Flexibility & Technical Pivot Adaptability
- Social Skills & Active Empathy with non-technical users
- Problem Solving
- Innovation & Systems Integration
- Critical Thinking

7. B.Sc. Software Engineering

EMPLOYABILITY SKILLS

Addresses the full lifecycle development, release, testing, maintenance, and scale of high-performance software systems:

- **Software Lifecycle Management:** Systems design, architectural modeling, systematic testing, and CI/CD pipelines.
- **Teamwork & Agile Collaboration:** Managing agile sprints, reviewing pull requests, and pairing code-level development.
- **Negotiation & Persuasion:** Working with product managers to scope requirements, features, and system milestones.
- **Problem-Solving:** Logical debugging, optimization of data structures, and multi-tier systems integration.
- **Organization & Perseverance:** Managing massive, long-term codebase scale and persevering through structural system refactors.

21ST CENTURY SKILLS

NUC Standard Skills

- Creative Thinking & Design Thinking
- Information Literacy & Documentation Mastery
- Media Literacy
- Flexibility & Tech-Stack Adaptability
- Social Skills & Pair Programming Synergy
- Complex Problem Solving
- Innovation Skills